

API Documentation

Overview

Base URL: `http://localhost:5000` (development) | `https://api.example.com` (production)

Response Format: JSON

Authentication: Bearer token

Rate Limits: 100 requests per minute

Authentication

Bearer Token

1. Obtain a token by logging in through the `/auth/login` endpoint with username and password.
2. Include in request headers: `Authorization: Bearer <token>`.
3. Token expires in 1 hour; refresh using the `/auth/refresh` endpoint.

Example Request with Auth

```
curl -X GET http://localhost:5000/api/leave-requests \
  -H "Authorization: Bearer your_token_here" \
  -H "Content-Type: application/json"
```

Endpoints

POST /api/leave-requests

Submit a new leave request.

Request:

- **Method:** POST
- **Path:** `/api/leave-requests`
- **Headers:** `Authorization: Bearer <token>`, `Content-Type: application/json`
- **Request Body:**

```
{
  "employeeId": "1",
  "leaveType": "SICK",
  "startDate": "2023-12-01",
  "endDate": "2023-12-05"
}
```

Response:

- **Status 201 (Created):**

```
{
  "success": true,
  "data": {
    "id": 1,
    "status": "PENDING"
  },
  "message": "Leave request submitted successfully"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 409 (Conflict):** Duplicate leave request

Example:

```
curl -X POST http://localhost:5000/api/leave-requests \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json" \
-d '{"employeeId": "1", "leaveType": "SICK", "startDate": "2023-12-01", "endDate": "2023-12-05"}
```

GET /api/leave-requests/{id}

Retrieve a leave request by ID.

Request:

- **Method:** GET
- **Path:** /api/leave-requests/{id}
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json

Response:

- **Status 200 (OK):**

```
{
  "success": true,
  "data": {
    "id": 1,
    "employeeId": 1,
    "leaveType": "SICK",
    "startDate": "2023-12-01",
    "endDate": "2023-12-05",
    "status": "PENDING"
  },
  "message": "Leave request retrieved successfully"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave request not found

Example:

```
curl -X GET http://localhost:5000/api/leave-requests/1 \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

GET /api/leave-requests

List all leave requests.

Request:

- **Method:** GET
- **Path:** /api/leave-requests
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json

Response:

- **Status 200 (OK):**

```
{
  "success": true,
  "data": [
    {
      "id": 1,
      "employeeId": 1,
      "leaveType": "SICK",
      "startDate": "2023-12-01",
      "endDate": "2023-12-05",
      "status": "PENDING"
    }
  ],
  "message": "Leave requests retrieved successfully"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions

Example:

```
curl -X GET http://localhost:5000/api/leave-requests \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

PUT /api/leave-requests/{id}/approve

Approve a leave request.

Request:

- **Method:** PUT
- **Path:** /api/leave-requests/{id}/approve
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json

Response:

- **Status 200 (OK):**

```
{
  "success": true,
  "data": {
    "id": 1,
    "status": "APPROVED"
  },
  "message": "Leave request approved successfully"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token

- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave request not found

Example:

```
curl -X PUT http://localhost:5000/api/leave-requests/1/approve \  
-H "Authorization: Bearer token" \  
-H "Content-Type: application/json"
```

PUT /api/leave-requests/{id}/reject

Reject a leave request.

Request:

- **Method:** PUT
- **Path:** /api/leave-requests/{id}/reject
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json

Response:

- **Status 200 (OK):**

```
{  
  "success": true,  
  "data": {  
    "id": 1,  
    "status": "REJECTED"  
  },  
  "message": "Leave request rejected successfully"  
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave request not found

Example:

```
curl -X PUT http://localhost:5000/api/leave-requests/1/reject \  
-H "Authorization: Bearer token" \  
-H "Content-Type: application/json"
```

DELETE /api/leave-requests/{id}

Cancel a leave request.

Request:

- **Method:** DELETE
- **Path:** /api/leave-requests/{id}
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json

Response:

- **Status 200 (OK):**

```
{
  "success": true,
  "data": {
    "id": 1,
    "status": "CANCELED"
  },
  "message": "Leave request canceled successfully"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave request not found

Example:

```
curl -X DELETE http://localhost:5000/api/leave-requests/1 \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

Error Codes and Meanings

Status	Code	Meaning	When It Occurs
400	BAD_REQUEST	Invalid input	Submitting a leave request with missing or invalid fields
401	UNAUTHORIZED	Invalid credentials	Token missing or expired
403	FORBIDDEN	Access denied	User lacks required role
404	NOT_FOUND	Resource not found	Leave request not found
409	CONFLICT	Duplicate or state conflict	Leave request already exists
429	RATE_LIMIT	Too many requests	Exceeded rate limit

Status	Code	Meaning	When It Occurs
500	INTERNAL_ERROR	Server error	Unexpected error, check logs

Rate Limits and Quotas

- **100 requests per minute:** Exceeding this limit will result in a 429 Too Many Requests response.

Data Types and Formats

- **Timestamps:** ISO 8601 format (e.g., 2026-01-15T10:30:00Z)
- **Booleans:** true or false
- **IDs:** UUID v4 or numeric, as documented per entity
- **Enums:** Specific values listed in request/response examples

Pagination (if applicable)

This API does not support pagination.

SDK Examples

No SDKs are provided for this API.